JAIPUR NATIONAL UNIVERSITY

NAAC

TECHNORAZZ 2026

HACKATHON

Hack the Gap

Problem Statement Title: Private Local-First Exam Proctoring Engine (PS 2.2)

Theme: EdTech /Privacy & Edge Security

Team Name: Edith

Team Number: 66

1 IDEA TITLE - EdgeProct: Proof Without Footage

Solution balances client-side processing with privacy.

Offline Resilience

Persists audit ledger in client IndexedDB

Tamper-Evident Ledger

Commits incidents to SHA-256 hash-chained log

Crypto-graphic Evidence

Replaces video upload with verifiable proof

Browser Integrity Guards

Intercepts suspicious browser activities



Explainable Fairness

Uses filters to avoid false flags

Dual Client-Side ML

Runs AI inferences on students device

Privacy-Pre-serving Data Transmission

Transmits signed event summaries, not raw video

Client-Side

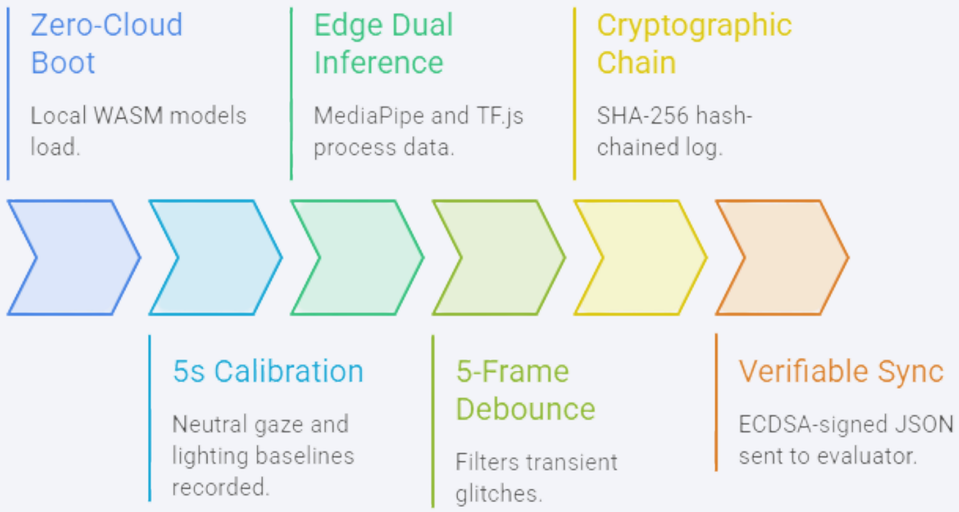
Server-Side

2 TECHNICAL APPROACH

1 Technologies to be Used

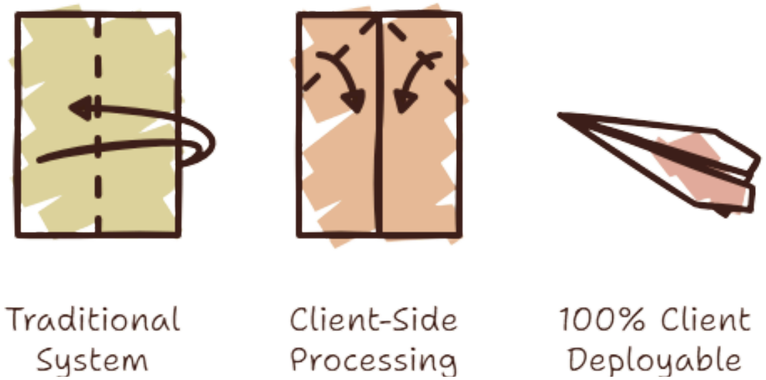
* **Frontend & Runtime:** React 19, Tailwind CSS, Vite, WebAssembly (WASM), WebGL.
* **Client Edge ML:** Google MediaPipe Tasks-Vision (Face Landmarker), TensorFlow.js COCO-SSD (bundled locally in /public for zero external CDN calls).
* **Cryptographic Security:** Web Crypto API (window.crypto.subtle) — SHA-256 hash chaining, ECDSA P-256 digital signatures.
* **Local Persistence:** HTML5 Web Storage, IndexedDB.

2 Methodology

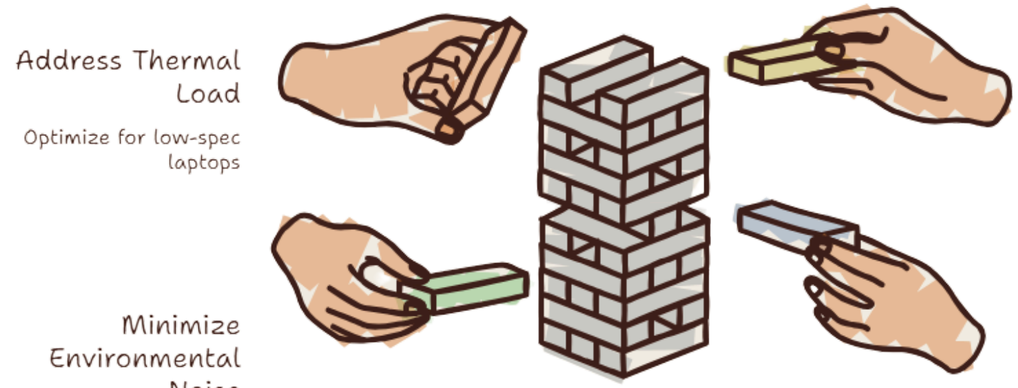


3 FEASIBILITY AND VIABILITY

Client-Side System Deployment



Overcoming Proctored Exam Challenges

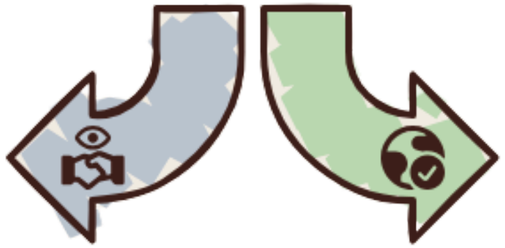


Mitigation Strategies

*__Adaptive Throttling:__ Caps processing at 2–3 FPS to keep CPU under 15%.
*__Device & Luma Checks:__ Detects virtual cameras and covered lenses via labels and brightness.
*__Smoothing Window:__ Filters noise by requiring violations in 3 out of 5 frames.
*__Hash Chaining:__ Uses SHA-256 to instantly expose any client-side log tampering.

4 IMPACT AND BENEFITS

What is the core impact of the remote examination solution?



Restores Digital Dignity and Trust

Enhances trust and dignity in remote examinations across various sectors.

Enables Standardized Assessments

Provides secure assessments in regions with low connectivity.

Which benefit of next-gen remote proctoring should be prioritized?



Social & Privacy

Ensures compliance with data protection laws and avoids intrusive surveillance.

Economic

Reduces costs by minimizing bandwidth and storage usage.

Technical

Maintains proctoring vigilance even with intermittent internet connectivity.

Empirical Metric

Provides significant data reduction compared to traditional methods.

5 RESEARCH AND REFERENCES

1 References

*Google MediaPipe & TensorFlow.js: Lugaresi et al., "MediaPipe: A Framework for Building Perception Pipelines" (In-browser Wasm & WebGL accelerated vision).
*Tamper-Evident Ledger Systems:S. A. Crosby & D. S. Wallach, "Efficient Data Structures for Tamper-Evident Logging" (Linear hash chains and cryptographic auditing).
*W3C Standards:W3C Web Cryptography API (SubtleCrypto P-256 ECDSA & SHA-256) and IndexedDB 3.0 specification.
*Regulatory & Privacy Frameworks:Ministry of Law and Justice, Digital Personal Data Protection Act (DPDP Act 2023), Government of India; EU General Data Protection Regulation (GDPR) Data Minimisation Principles.